

# Lesson 9: Grover Search and Amplitude Amplification

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Oracle, diffusion operator, interference, and the marked state

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# Learning Outcomes

- Explain the search problem in Grover's algorithm.
- Distinguish an oracle from a diffusion/amplification step.
- Trace the two-qubit circuit in the notebook.
- Interpret why 11 dominates the measurement result.

# The Search Problem

- There are four basis states for two qubits: 00, 01, 10 and 11.
- The oracle marks the desired state.
- The algorithm amplifies the marked state amplitude.
- Measurement is then likely to return the marked state.

# Oracle vs Diffusion

## Oracle

- Marks  $|11\rangle$ .
- Uses Controlled-Z.

## Diffusion

- Repositions amplitudes.
- Uses H, X, CZ, X, H.

# Read the Grover Code

```
def grover_oracle():  
    oracle = QuantumCircuit(2)  
    oracle.cz(0, 1)  
    return oracle  
  
def diffusion_operator():  
    diffusion = QuantumCircuit(2)  
    diffusion.h([0, 1])  
    diffusion.x([0, 1])  
    diffusion.cz(0, 1)  
    diffusion.x([0, 1])  
    diffusion.h([0, 1])  
    return diffusion  
  
qc = QuantumCircuit(2, 2)  
qc.h([0, 1])  
qc = qc.compose(grover_oracle())  
qc = qc.compose(diffusion_operator())  
qc.measure([0, 1], [0, 1])  
  
simulator = Aer.get_backend('aer_simulator')
```

## Interpreting the Recorded Output

$$\{11: 1000\}$$

- Every one of the 1000 recorded shots is 11 in the notebook output.
- For this two-qubit example, the marked state is therefore strongly amplified.
- The result illustrates the idea of quantum amplitude amplification.

# Why This Is Not Brute Force

- A classical search checks candidates explicitly.
- Grover uses quantum interference to amplify the desired amplitude.
- For larger search spaces, Grover provides a quadratic query-speedup in the idealized oracle model.

# Beginner Activity

- Change the oracle so a different state is marked.
- Predict which bit string should dominate.
- Run the circuit and compare the histogram.

- The oracle identifies the desired state.
- The diffusion step amplifies its amplitude.
- The notebook obtains 11 in all 1000 recorded shots.

**Grover shows how interference can turn a broad superposition into a concentrated answer.**

## Quick Check

- ❶ What does the oracle do?
- ❷ What does diffusion do?
- ❸ Why are Hadamard gates used at the beginning?